

Fuzzy Logic Programming and Fuzzy Control

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Abstract. We show that it is possible to base fuzzy control on fuzzy logic programming. Indeed, we observe that the class of fuzzy Herbrand interpretations gives a semantics for fuzzy programs and we show that the fuzzy function associated with a fuzzy system of IF-THEN rules is the fuzzy Herbrand interpretation associated with a suitable fuzzy program.

Keywords: Fuzzy control, fuzzy logic programming, Herbrand interpretation.

1. Introduction and preliminaries

The basic ideas of fuzzy logic (in narrow sense) were formulated by L. Zadeh (see [19], [20]), J. A. Goguen (see [9]) and J. Pavelka (see [14], [15], and [16]). The aim of such a logic is to formalize the "approximate reasoning" we use in everyday life and this by admitting predicates as "*big*", "*near*", "*slow*" which are vague in nature. Also, these predicate are interpreted by suitable "fuzzy subsets", i.e. generalized characteristic functions with values in a complete lattice. Fuzzy logic programming is a very promising chapter of fuzzy logic whose purpose is to build up intelligent data-base systems with "flexible" answers, expert systems able to consider vague predicates and so on (see, for example, [5], [2], [13], [17] and [18]).

In this paper at first we recall some basic notions in fuzzy logic programming, which are well known in literature. Also, we show that is not restrictive to confine ourselves to fuzzy Herbrand interpretations in giving a semantics for fuzzy programs. This is done by showing that any homomorphism preserves the universal formulas. Moreover, by developing some ideas sketched in [8], we relate fuzzy logic programming with fuzzy control (see also P. Hájek in [10]). This gives a unifying and rigorous framework for both conjunction-based and implication-based fuzzy control.

Let L be a complete, completely distributive, lattice whose elements we call *truth values* and let S be a nonempty set. Then, an L -subset or *fuzzy subset* of S is any map $s : S \rightarrow L$ from S to L . Given $x \in S$, we say that the value $s(x)$ is the membership degree of x to s . For any $\lambda \in \mathbf{L}$, the

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set $C(s, \lambda) = \{x \in S : s(x) \geq \lambda\}$ is called the λ -cut of s . The subset $\text{Supp}(s) = \{x \in S : s(x) \neq 0\}$ is called the support of s . If 0 denotes the minimum and 1 the maximum of L , then we call *crisp* any L -subset whose values are in $\{0, 1\}$. We can identify any subset X of S with the *crisp* subset $c_X : S \rightarrow \{0, 1\}$ such that $c_X(x) = 1$ if $x \in X$ and $c_X(x) = 0$ in the case $x \notin X$. In particular, we identify $\emptyset$ with the map constantly equal to 0 and S with the map constantly equal to 1. We say that a fuzzy subset s is contained in a fuzzy subset s' provided that $s(x) \leq s'(x)$ for any $x \in S$. In such a case, we write $s \subseteq s'$. Given a family $(s_i)_{i \in I}$ of fuzzy subsets, the union and the intersection are defined by setting

$$\bigcup_{i \in I} s_i(x) = \text{Sup}\{s_i(x) : i \in I\} \quad ; \quad \bigcap_{i \in I} s_i(x) = \text{Inf}\{s_i(x) : i \in I\};$$

for any $x \in S$. With respect to these operations, the class $\mathfrak{F}(S)$ of all the L -subsets of S is a complete lattice, i.e. the direct power of L with index set S . Let $S_1, \dots, S_n$ be sets, then an n -ary L -relation on $S_1, \dots, S_n$ is any L -subset of the Cartesian product $S_1 \times \dots \times S_n$. A binary L -relation $f : S_1 \times S_2 \rightarrow L$ is also called (*non-deterministic*) L -function from S_1 to S_2 . The idea is that, given the input $x \in S_1$, the output is not an element in S_2 but the fuzzy subset $s : S_2 \rightarrow L$ of elements of S_2 defined by setting $s(y) = f(x, y)$ for any $y \in S_2$. Let $\otimes$ be a binary operation in L and $a : S_1 \rightarrow L$ and $b : S_2 \rightarrow L$ two fuzzy subsets of S_1 and S_2 , respectively. Then the $\otimes$ -Cartesian product $a \times b : S_1 \times S_2 \rightarrow L$ is defined by setting, for any $(x, y) \in S_1 \times S_2$,

$$(a \times b)(x, y) = a(x) \otimes b(y).$$

2. Fuzzy interpretations of a first order language

Let $\mathcal{L}$ be a first order language defined, as usual, by:

- a set LC of logical connectives,
- a set OS of operation symbols,
- the universal quantifier $\forall$,
- a set RS of relation symbols,
- a set C of constants,
- a map $ar : (RS \cup OS \cup LC) \rightarrow N$.

We consider also a sequence $x_1, x_2, \dots$ of symbols that we call *variables* and the brackets $(,)$. If s is a symbol such that $ar(s) = n$, then we say that s is n -ary. Terms and formulas are defined as in classic first order logic. Namely, we assume that:

- the constants and the variables are terms,
- if h is an p -ary operation symbol and $t_1, \dots, t_p$ are terms, then $h(t_1, \dots, t_p)$ is a term.

An *atomic formula* is an expression like $r(t_1, \dots, t_p)$, where r is a p -ary relation symbol and $t_1, \dots, t_p$ are terms. The whole set F of formulas is defined by assuming that:

- each atomic formula is a formula,
- if l is an k -ary logical connective and $\alpha_1, \dots, \alpha_k$ are formulas, then $l(\alpha_1, \dots, \alpha_k)$ is a formula,
- if α is a formula and x_i a variable, then $\forall x_i(\alpha)$ is a formula.

A term (a formula) is *ground* if no variable occurs in it. A ground atom is called a *fact*. Then a ground formula is obtained from facts by the logical connectives. The notions of *free occurrence* of a variable in a formula, of *quantifier-free* formula, of *universal formula* and so on, are defined as usual. If s is either a term or a formula, we write $s(x_1, \dots, x_n)$ to emphasize that the variables occurring in s are among $x_1, \dots, x_n$.

We will define the semantics by taking in account the possibility that some relation symbols represent vague predicates that we model by the notion of L -subset.

DEFINITION 2.1. An L -interpretation or fuzzy interpretation of $\mathfrak{L}$ is a pair (D, I) , where D is a nonempty set and I is a map associating

- any n -ary function symbol h with a function $I(h) : D^n \rightarrow D$,
- any constant c with an element $I(c) \in D$
- any n -ary relation symbol r with an L -relation $I(r) : D^n \rightarrow L$.

We call *domain* the set D and sometimes we write I to denote (D, I) . The terms are interpreted as in the classical case.

DEFINITION 2.2. Given a term $t(x_1, \dots, x_n)$, the *the interpretation* of t in (D, I) is the map $I(t) : D^n \rightarrow D$ defined by setting:

- $I(x_i)(d_1, \dots, d_n) = d_i$,
- $I(c)(d_1, \dots, d_n) = I(c)$,
- $I(h(t_1, \dots, t_k))(d_1, \dots, d_n) = I(h)(I(t_1)(d_1, \dots, d_n), \dots, I(t_k)(d_1, \dots, d_n))$.

We assume that any n -ary logical connective $l \in LC$ is associated with an n -ary operation $\underline{l} : L^n \rightarrow L$ and this enables us to associate any formula with the related extension.

DEFINITION 2.3. Given a formula $\alpha(x_1, \dots, x_n)$, the *extension* $I(\alpha) : D^n \rightarrow L$ of α is the n -ary fuzzy relation defined by setting:

- $I(r(t_1, \dots, t_p))(d_1, \dots, d_n) = I(r)(I(t_1)(d_1, \dots, d_n), \dots, I(t_p)(d_1, \dots, d_n))$,
- $I(l(\alpha_1, \dots, \alpha_k))(d_1, \dots, d_n) = \underline{l}(I(\alpha_1)(d_1, \dots, d_n), \dots, I(\alpha_k)(d_1, \dots, d_n))$,
- $I(\forall x_i(\alpha))(d_1, \dots, d_n) = \bigwedge_{d \in D} I(\alpha)(d_1, \dots, d_{i-1}, d, d_{i+1}, \dots, d_n)$.

Trivially, if α is a closed formula, then $I(\alpha)$ is a constant map, i.e. the quantity $I(\alpha)(d_1, \dots, d_n)$ does not depend on the elements $d_1, \dots, d_n$. In such a case we denote such a constant again by $I(\alpha)$.

DEFINITION 2.4. Let α be any formula and $\forall x_1 \dots \forall x_n(\alpha)$ be its universal closure. Then we set

$$\|\alpha\|_I = I(\alpha)(\forall x_1 \dots \forall x_n(\alpha)).$$

As an obvious extension of the notion of set of axioms in classical logic, we call *fuzzy set of axioms* or *fuzzy theory* any fuzzy set $\tau : F \rightarrow L$ of formulas.

DEFINITION 2.5. Let $\tau : F \rightarrow L$ be a fuzzy set of axioms. Then an L -interpretation (D, I) is called a *model of τ* (for short, $I \models \tau$), if $\|\varphi\|_I \geq \tau(\varphi)$ for any $\varphi \in F$.

In accordance with such a definition, the value $\tau(\varphi)$ is not intended as the truth value of φ but as a lower-bound constraint on the possible truth value of φ . Then, the information carried on by a fuzzy set of axioms τ is that, for any formula φ , the truth value of φ is greater than or equal to $\tau(\varphi)$.

DEFINITION 2.6. Let $\tau : F \rightarrow L$ be a fuzzy set of axioms. Then the *fuzzy set $Lc(\tau)$ of logical consequences of τ* is defined by setting:

$$LC(\tau)(\varphi) = Inf\{\|\varphi\|_I : I \models \tau\}. \quad (2.1)$$

The so defined operator $Lc : \mathfrak{F}(F) \rightarrow \mathfrak{F}(F)$ is called the *logical consequence operator*. In a sense, $LC(\tau)(\varphi)$ is the best lower-bound constraint on the truth value of φ , we can find given the available information τ .

3. Homomorphisms and preserving theorems

Given two L -interpretations (D, I) and (D', I') , we call *homomorphism* from (D, I) into (D', I') any map $f : D \rightarrow D'$ such that:

- f is a homomorphism of the associated algebraic structures, i.e.,

$$f(I(c)) = I'(c) \quad ; \quad f(I(h)(d_1, \dots, d_n)) = I'(h)(f(d_1), \dots, f(d_n)),$$
for any $c \in C$ any n -ary operation symbol h and $d_1, \dots, d_n \in D$,
- for every n -ary relation r ,

$$I(r)(d_1, \dots, d_n) \leq I'(r)(f(d_1), \dots, f(d_n)). \quad (3.1)$$

We say that f is *full* if

$$I(r)(d_1, \dots, d_n) = I'(r)(f(d_1), \dots, f(d_n)). \quad (3.2)$$

An *isomorphism* is a homomorphism which is one-to-one and full, i.e. an one-to-one homomorphism whose inverse is again a homomorphism. Given an interpretation (D, I) and $\lambda \in L$, we call λ -cut of I the classical interpretation (D, I_λ) defined by setting $I_\lambda(r) = C(I(r), \lambda)$ for any relation symbol r , $I_\lambda(c) = I(c)$ for any $c \in C$ and $I_\lambda(h) = I(h)$ for any operation symbol h .

PROPOSITION 3.1. Let (D, I) and (D', I') be two L -interpretations and $f : D \rightarrow D'$ be a map. Then f is a homomorphism (full homomorphism, isomorphism) from (D, I) into (D', I') if and only if f is a homomorphism (full homomorphism, isomorphism) from (D, I_λ) into (D', I'_λ) for any $\lambda \in L$.

PROOF. Assume that f is a homomorphism from (D, I) to (D', I') . Then, for any $\lambda \in L$ and for any n -ary relation symbol r ,

$$\begin{aligned} (d_1, \dots, d_n) \in C(I(r), \lambda) &\Leftrightarrow I(r)(d_1, \dots, d_n) \geq \lambda \\ &\Rightarrow I'(r)(f(d_1), \dots, f(d_n)) \geq I(r)(d_1, \dots, d_n) \geq \lambda \\ &\Leftrightarrow (f(d_1), \dots, f(d_n)) \in C(I'(r), \lambda). \end{aligned}$$

This proves that f is a homomorphism from (D, I_λ) into (D', I'_λ) . Conversely, assume that, for any $\lambda \in L$, f is a homomorphism from (D, I_λ) into (D', I'_λ) and therefore that $I(r)(d_1, \dots, d_n) \geq \lambda$ entails $I'(r)(f(d_1), \dots, f(d_n)) \geq \lambda$. Then, by setting $\lambda = I(r)(d_1, \dots, d_n)$, we obtain that $I'(r)(f(d_1), \dots, f(d_n)) \geq I(r)(d_1, \dots, d_n)$. This proves that f is a homomorphism from (D, I) into (D', I') . The remaining part of the proof is trivial. ■

As in classical logic, given a homomorphism f and a term $t(x_1, \dots, x_n)$, we have that for every $d_1, \dots, d_n \in D$,

$$f(I(t)(d_1, \dots, d_n)) = I'(t)(f(d_1), \dots, f(d_n)). \quad (3.3)$$

THEOREM 3.2. Let $\varphi(x_1, \dots, x_n)$ a quantifier-free formula, (D, I) and (D', I') be two fuzzy interpretations, and $f : D \rightarrow D'$ be a full homomorphism. Then, for any $d_1, \dots, d_n \in D$,

$$I(\varphi)(d_1, \dots, d_n) = I'(\varphi)(f(d_1), \dots, f(d_n)). \quad (3.4)$$

If $\psi \in F$ is universal,

$$I(\psi)(d_1, \dots, d_n) \geq I'(\psi)(f(d_1), \dots, f(d_n)) \quad (3.5)$$

and therefore

$$\|\psi\|_I \geq \|\psi\|_{I'}. \quad (3.6)$$

PROOF. We will prove (3.4) by induction on the complexity of φ . Let φ be the atomic formula $r(t_1, \dots, t_p)$. Then, by (3.2) and (3.3),

$$\begin{aligned} I(\varphi)(d_1, \dots, d_n) &= I(r)(I(t_1)(d_1, \dots, d_n), \dots, I(t_p)(d_1, \dots, d_n)) \\ &= I'(r)(f(I(t_1)(d_1, \dots, d_n)), \dots, f(I(t_p)(d_1, \dots, d_n))) \\ &= I'(r)(I'(t_1)(f(d_1), \dots, f(d_n))), \dots, I'(t_p)(f(d_1), \dots, f(d_n))) \\ &= I'(\varphi)(f(d_1), \dots, f(d_n)). \end{aligned}$$

Assume that $\varphi = l(\alpha_1, \dots, \alpha_k)$ and that (3.4) is satisfied by $\alpha_1, \dots, \alpha_k$. Then,

$$\begin{aligned}
I(\varphi)(d_1, \dots, d_n) &= l(I(\alpha_1)(d_1, \dots, d_n), \dots, I(\alpha_k)(d_1, \dots, d_n)) \\
&= l(I'(\alpha_1)(f(d_1), \dots, f(d_n)), \dots, I'(\alpha_k)(f(d_1), \dots, f(d_n))) \\
&= I'(\varphi)(f(d_1), \dots, f(d_n)).
\end{aligned}$$

This proves the first part of the theorem. Let ψ be universal, we will prove (3.5) by induction on the number m of universal quantifiers in ψ . If $m = 0$, (3.5) follows from (3.4). Assume that $m \neq 0$ and that ψ is equal to a formula like $\forall x_i(\alpha)$, where α contains $m-1$ universal quantifiers. Then, in accordance with the induction hypothesis and the inclusion $\{f(d) : d \in D\} \subseteq D'$,

$$\begin{aligned}
I(\forall x_i(\alpha))(d_1, \dots, d_n) &= \text{Inf}_{d \in D} I(\alpha)(d_1, \dots, d, \dots, d_n) \\
&\geq \text{Inf}_{d \in D} I'(\alpha)(f(d_1), \dots, f(d), \dots, f(d_n)) \\
&\geq \text{Inf}_{d' \in D'} I'(\alpha)(f(d_1), \dots, d', \dots, f(d_n)) \\
&= I'(\forall x_i(\alpha))(f(d_1), \dots, f(d_n)). \quad \blacksquare
\end{aligned}$$

Not surprisingly, the following theorem proves that two isomorphic fuzzy interpretations are logically equivalent.

THEOREM 3.3. *Let f be an isomorphism between (D, I) and (D', I') . Then for any formula α , we have that*

$$I(\alpha)(d_1, \dots, d_n) = I'(\alpha)(f(d_1), \dots, f(d_n)). \quad (3.7)$$

PROOF. We prove (3.7) by induction on the complexity of α . If α is atomic, (3.7) was just proved in Theorem 3.2. Again, in proving Theorem 3.2 we showed that if l is a logical connective and $\alpha_1, \dots, \alpha_k$ satisfy (3.7), then $l(\alpha_1, \dots, \alpha_k)$ satisfies (3.7). Assume that α satisfies (3.7). Then, since $\{f(d) : d \in D\} = D'$,

$$\begin{aligned}
I(\forall x_i \alpha)(d_1, \dots, d_n) &= \text{Inf}_{d \in D} I(\alpha)(d_1, \dots, d, \dots, d_n) \\
&= \text{Inf}_{d \in D} I'(\alpha)(f(d_1), \dots, f(d), \dots, f(d_n)) \\
&= \text{Inf}_{d' \in D'} I'(\alpha)(f(d_1), \dots, d', \dots, f(d_n)) \\
&= I'(\forall x_i \alpha)(f(d_1), \dots, f(d_n)).
\end{aligned}$$

Thus, (3.7) is satisfied by any formula. ■

4. Fuzzy Herbrand interpretations and universal theories

As usual, we call *Herbrand universe* the set $\mathfrak{U}$ of ground terms of the language $\mathcal{L}$ and *Herbrand base* the set $B_{\mathcal{L}}$ of facts.

DEFINITION 4.1. An L -interpretation (D, I) is an L -Herbrand interpretation (for short H -interpretation) provided that:

- D is equal to the Herbrand universe $\mathfrak{U}$,
- for any n -ary operation symbol h , and $t_1, \dots, t_n \in \mathfrak{U}$,

$$I(h)(t_1, \dots, t_n) = h(t_1, \dots, t_n),$$

- for any constant c , $I(c) = c$.

Then, two H -interpretations differ only for the relational part and this enables us to identify the H -interpretations with the fuzzy subsets of $B_{\mathcal{L}}$.

PROPOSITION 4.2. There is a one-one-mapping between the class of H -interpretations and the class of fuzzy subsets of $B_{\mathcal{L}}$.

PROOF. We associate any H -interpretation $(\mathfrak{U}, \mathfrak{I})$ with the fuzzy subset $s_{\mathfrak{I}}$ of $B_{\mathcal{L}}$ defined by setting, $s_{\mathfrak{I}}(r(t_1, \dots, t_p)) = \mathfrak{I}(r(t_1, \dots, t_p))$, for any fact $r(t_1, \dots, t_p)$. Moreover, we associate any fuzzy subset s of $B_{\mathcal{L}}$ with the H -interpretation $(\mathfrak{U}, \mathfrak{I}_s)$ where, for any p -ary relation symbol r , $\mathfrak{I}_s(r)$ is defined by setting, for any $t_1, \dots, t_p$ in $\mathfrak{U}$, $\mathfrak{I}_s(r)(t_1, \dots, t_p) = s(r(t_1, \dots, t_p))$. ■

In account of such a proposition, in the following we call H -interpretation any fuzzy subset of $B_{\mathcal{L}}$. In order to prove some basic properties of the H -interpretations, we need some further notations. Namely, let $t(x_1, \dots, x_n)$ be a term. Then, given the terms $t_1, \dots, t_n$, we write $t(t_1, \dots, t_n)$ to denote the term obtained by substituting each x_i with t_i in $t(x_1, \dots, x_n)$. Likewise, given the formula $\alpha(x_1, \dots, x_n)$, $\alpha(t_1, \dots, t_n)$ denotes the formula obtained from $\alpha(x_1, \dots, x_n)$ by substituting each variable x_i which is free in α with t_i . Trivially, if t is ground, then $t(t_1, \dots, t_n)$ is equal to t , if α is closed, then $\alpha(t_1, \dots, t_n)$ coincides with α .

PROPOSITION 4.3. Let $(\mathfrak{U}, \mathfrak{I})$ be an H -interpretation, $t(x_1, \dots, x_n)$ a term and $t_1, \dots, t_n \in \mathfrak{U}$. Then

$$\mathfrak{I}(t)(t_1, \dots, t_n) = t(t_1, \dots, t_n). \quad (4.1)$$

Moreover, if $\alpha(x_1, \dots, x_n)$ is a formula,

$$\mathfrak{I}(\alpha)(t_1, \dots, t_n) = \|\alpha(t_1, \dots, t_n)\|_{\mathfrak{I}}. \quad (4.2)$$

Consequently,

$$\|\forall x_1 \dots \forall x_n (\alpha)\|_{\mathfrak{I}} = \text{Inf}\{\|\alpha(t_1, \dots, t_n)\|_{\mathfrak{I}} : t_1, \dots, t_n \in \mathfrak{U}\}. \quad (4.3)$$

PROOF. (4.1) one proves as in the classical logic. We prove (4.2) by induction on the complexity of α . Indeed, assume that α is the atomic formula $r(s_1, \dots, s_p)$. Then

$$\begin{aligned}
\mathfrak{I}(\alpha)(t_1, \dots, t_n) &= \mathfrak{I}(r)(\mathfrak{I}(s_1)(t_1, \dots, t_n), \dots, \mathfrak{I}(s_p)(t_1, \dots, t_n)) \\
&= \mathfrak{I}(r)(s_1(t_1, \dots, t_n), \dots, s_m(t_1, \dots, t_n)) \\
&= \mathfrak{I}(r(s_1(t_1, \dots, t_n), \dots, s_m(t_1, \dots, t_n))) \\
&= \|\alpha(t_1, \dots, t_n)\|_{\mathfrak{I}}.
\end{aligned}$$

Assume that $\alpha = l(\alpha_1, \dots, \alpha_k)$ and that (4.2) is satisfied by each α_i , then

$$\begin{aligned}
\mathfrak{I}(l(\alpha_1, \dots, \alpha_k))(t_1, \dots, t_n) &= \underline{l}(\mathfrak{I}(\alpha_1)(t_1, \dots, t_n), \dots, \mathfrak{I}(\alpha_k)(t_1, \dots, t_n)) \\
&= \underline{l}(\|\alpha_1(t_1, \dots, t_n)\|_{\mathfrak{I}}, \dots, \|\alpha_k(t_1, \dots, t_n)\|_{\mathfrak{I}}) \\
&= \|l(\alpha_1, \dots, \alpha_k)(t_1, \dots, t_n)\|_{\mathfrak{I}}.
\end{aligned}$$

This proves that (4.2) is satisfied by $l(\alpha_1, \dots, \alpha_k)$. Finally, assume that α is the formula $\forall x_i(\beta)$ and that, by induction hypothesis, for any $t \in \mathfrak{U}$, $\mathfrak{I}(\beta)(t_1, \dots, t, \dots, t_n) = \|\beta(t_1, \dots, t, \dots, t_n)\|_{\mathfrak{I}}$ and $\mathfrak{I}(\beta(t_1, \dots, x_i, \dots, t_n))(t) = \|\beta(t_1, \dots, t, \dots, t_n)\|_{\mathfrak{I}}$. Then, since $\forall x_i(\beta(t_1, \dots, x_i, \dots, t_n))$ is equal to the formula $(\forall x_i(\beta))(t_1, \dots, t_n)$, we get:

$$\begin{aligned}
\mathfrak{I}(\forall x_i(\beta))(t_1, \dots, t_n) &= \text{Inf}\{\mathfrak{I}(\beta)(t_1, \dots, t, \dots, t_n) : t \in \mathfrak{U}\} \\
&= \text{Inf}\{\|\beta(t_1, \dots, t, \dots, t_n)\|_{\mathfrak{I}} : t \in \mathfrak{U}\} \\
&= \text{Inf}\{\mathfrak{I}(\beta(t_1, \dots, x_i, \dots, t_n))(t) : t \in \mathfrak{U}\} \\
&= \|\forall x_i(\beta(t_1, \dots, x_i, \dots, t_n))\|_{\mathfrak{I}} \\
&= \|(\forall x_i(\beta))(t_1, \dots, t_n)\|_{\mathfrak{I}}.
\end{aligned}$$

This proves that (4.2) is satisfied by the formula $\forall x_i(\beta)$, too. ■

Given an L -interpretation (D, I) , the H -interpretation associated with (D, I) is the H -interpretation $(\mathfrak{U}, \mathfrak{I})$ defined by the fuzzy set of facts which are true in (D, I) . This means that, for any n -ary relation symbol r , it is

$$\mathfrak{I}(r)(t_1, \dots, t_n) = I(r)(I(t_1), \dots, I(t_n)) = I(r(t_1, \dots, t_n)),$$

for any $t_1, \dots, t_n \in \mathfrak{U}$. The proof of the following proposition is trivial.

PROPOSITION 4.4. Let (D, I) be an L -interpretation and $(\mathfrak{U}, \mathfrak{I})$ be the associated H -interpretation. Then the map $f : \mathfrak{U} \rightarrow D$, defined by setting $f(t) = I(t)$ for any $t \in \mathfrak{U}$, is a full homomorphism from $\mathfrak{I}$ into I . Consequently, if φ is a quantifier-free formula, then for any $t_1, \dots, t_n \in \mathfrak{U}$:

$$\mathfrak{I}(\varphi)(t_1, \dots, t_n) = I(\varphi)(I(t_1), \dots, I(t_n)). \quad (4.4)$$

If ψ is universal,

$$\mathfrak{I}(\psi)(t_1, \dots, t_n) \geq I(\psi)(I(t_1), \dots, I(t_n)), \quad (4.5)$$

and therefore

$$\|\psi\|_{\mathfrak{I}} \geq \|\psi\|_I. \quad (4.6)$$

We say that a fuzzy theory $\tau : F \rightarrow L$ is *universal* if all the formulas in $Supp(\tau)$ are universal. From (4.6) we obtain the following proposition:

PROPOSITION 4.5. Let τ be an universal fuzzy theory, (D, I) be an L -interpretation and $(\mathfrak{U}, \mathfrak{J})$ the associated H -interpretation. Then

$$(D, I) \text{ is a model of } \tau \Rightarrow (\mathfrak{U}, \mathfrak{J}) \text{ is a model of } \tau.$$

The following proposition shows that we can define the fuzzy set of facts which are logical consequences of an universal fuzzy theory, by referring only to the fuzzy Herbrand models of this theory.

PROPOSITION 4.6. Let τ be an universal fuzzy theory. Then, for any fact φ ,

$$Lc(\tau)(\varphi) = Inf\{\|\varphi\|_{\mathfrak{J}} : \mathfrak{J} \text{ is a } H\text{-model of } \tau\}. \quad (4.7)$$

PROOF. Let φ be a fact. Then, since by (4.4), $\|\varphi\|_{\mathfrak{J}} = \|\varphi\|_I$,

$$\begin{aligned} Lc(a)(\varphi) &= Inf\{\|\varphi\|_I : I \text{ is a model of } \tau\} \\ &= Inf\{\|\varphi\|_{\mathfrak{J}} : \mathfrak{J} \text{ is associated with a model of } \tau\} \\ &= Inf\{\|\varphi\|_{\mathfrak{J}} : \mathfrak{J} \text{ is a } H\text{-model of } \tau\}. \quad \blacksquare \end{aligned}$$

Let $\tau : F \rightarrow L$ be a universal fuzzy theory. Then we define the *fuzzy set* $Gr(\tau)$ of ground instances of the formulas in τ by setting, $Gr(\tau)(\alpha) = 0$ in the case α is not ground and

$$Gr(\tau)(\alpha) = Sup\{\tau(\forall x_1 \dots \forall x_n(\alpha')) : \alpha = \alpha'(t_1, \dots, t_n), t_1, \dots, t_n \in \mathfrak{U}\},$$

otherwise. We will show that if τ is an universal theory, then τ and $Gr(\tau)$ have the same H -models.

THEOREM 4.7. Let τ be an universal fuzzy theory, then, for any H -interpretation I ,

$$I \models \tau \Leftrightarrow I \models Gr(\tau).$$

PROOF. Assume that $I \models \tau$ and let α be a formula. Then, if α is not ground, $Gr(\tau)(\alpha) = 0$ and therefore $\|\alpha\|_I \geq Gr(\tau)(\alpha)$. Assume that α is ground and let α' be a formula such that $\alpha = \alpha'(t_1, \dots, t_n)$ for some $t_1, \dots, t_n \in \mathfrak{U}$. Then

$$\|\alpha\|_I \geq \|\forall x_1 \dots \forall x_n(\alpha')\|_I \geq \tau(\forall x_1 \dots \forall x_n(\alpha')).$$

Consequently,

$$\|\alpha\|_I \geq Sup\{\tau(\forall x_1 \dots \forall x_n(\alpha')) : \alpha = \alpha'(t_1, \dots, t_n), t_1, \dots, t_n \in \mathfrak{U}\} = Gr(\tau)(\alpha),$$

and this proves that I is a model of $Gr(\tau)$. Conversely, let I be a H -model of $Gr(\tau)$ and let $\forall x_1 \dots \forall x_n(\alpha')$ be any closed universal formula. Then, in accordance with the definition of $Gr(\tau)$, given $t_1, \dots, t_n$ in $\mathfrak{U}$,

$$\|\alpha'(t_1, \dots, t_n)\|_I \geq Gr(\tau)(\alpha'(t_1, \dots, t_n)) \geq \tau(\forall x_1 \dots \forall x_n(\alpha')).$$

Thus

$\|\forall x_1 \dots \forall x_n (\alpha')\|_I = \text{Inf}\{\|\alpha'(t_1, \dots, t_n)\|_I : t_1, \dots, t_n \in \mathfrak{U}\} \geq \tau(\forall x_1 \dots \forall x_n (\alpha'))$
and this proves that I is a model of τ . ■

5. Fuzzy logic programming

Now we will expose some definitions and results in fuzzy logic programming which are well known in literature (see, for example, [2], [13], [17] and [18]). Assume that in L there are some commutative and associative operations $\otimes_1, \dots, \otimes_k$ such that, for any x in L , $x \otimes_k 1 = x$ and, for any family $(y_j)_{j \in J}$ of elements in L ,

$$x \otimes_i (\text{Sup}_{j \in J} y_j) = \text{Sup}_{j \in J} (x \otimes_i y_j).$$

Under these conditions, we can define the implication operations $\rightarrow_i$ by setting, for any $x, y \in L$,

$$x \rightarrow_i y = \text{Sup}\{z \in L : x \otimes_i z \models y\}.$$

It is easy to prove that each pair $\rightarrow_i$ and $\otimes_i$ satisfies the *adjunction property*, i.e, for any $a, b, x \in L$,

$$a \rightarrow_i b \geq x \Leftrightarrow b \geq x \otimes_i a.$$

We suppose that $\mathfrak{L}$ contains logical connectives corresponding to these operations we denote again by $\rightarrow_1, \dots, \rightarrow_k$. Let $\mathfrak{L}^*$ be the sentential calculus with the same logical connectives as in $\mathfrak{L}$. Then we write $h(p_1, \dots, p_n)$ to denote a formula of $\mathfrak{L}^*$ whose propositional variables are among $p_1, \dots, p_n$ and, as usual, we associate $h(p_1, \dots, p_n)$ with a *truth-table* $\underline{h} : L^n \rightarrow L$. Let $\alpha_1, \dots, \alpha_n$ be atomic formulas of $\mathfrak{L}$ and $h(\alpha_1, \dots, \alpha_n)$ be the formula in $\mathfrak{L}$ obtained from $h(p_1, \dots, p_n)$ by substituting each occurrence of p_i with α_i . Then we say that $h(\alpha_1, \dots, \alpha_n)$ is a *combination* of the formulas $\alpha_1, \dots, \alpha_n$. If $\underline{h}$ preserves the least upper bounds (and therefore is monotone), then we say that $h(\alpha_1, \dots, \alpha_n)$ is a *positive combination* of $\alpha_1, \dots, \alpha_n$.

DEFINITION 5.1. A *(positive) implicative clause* is either an atomic formula or a formula like $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ where $\alpha, \alpha_1, \dots, \alpha_n$ are atomic formulas and $h(\alpha_1, \dots, \alpha_n)$ is a (positive) combination of $\alpha_1, \dots, \alpha_n$. A fuzzy subset $p : F \rightarrow L$ of formulas is a *(positive, ground) fuzzy program* if $\text{Supp}(p)$ is a set of (positive, ground) implicative clauses.

A simple characterization of the fuzzy models of a fuzzy program is the following.

PROPOSITION 5.2. Let $p : F \rightarrow L$ be a fuzzy program and (D, I) be an interpretation. Then I is a fuzzy model of p if and only if, for any $d_1, \dots, d_h$ in D ,

- i) $I(\alpha)(d_1, \dots, d_h) \geq p(\alpha)$
for any atomic formula $\alpha \in \text{Supp}(p)$;

- ii) $I(\alpha)(d_1, \dots, d_h) \geq p(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha)$
 $\otimes_i \underline{h}(I(\alpha_1)(d_1, \dots, d_h), \dots, I(\alpha_n)(d_1, \dots, d_h))$
 for any implicative clause $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ in $\text{Supp}(p)$.

PROOF. An interpretation I is a model of p if and only if,

$$\|\alpha\|_I = \text{Inf}\{I(\alpha)(d_1, \dots, d_h) : d_1, \dots, d_h \in D\} \geq p(\alpha)$$

for any atomic formula $\alpha \in \text{Supp}(p)$ and

$$\|h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha\|_I = \text{Inf}\{I(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha)(d_1, \dots, d_h) : d_1, \dots, d_h \in D\} \\ \geq p(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha),$$

for any formula $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ in $\text{Supp}(p)$. The first condition is equivalent to the claim that $I(\alpha)(d_1, \dots, d_h) \geq p(\alpha)$ for any $d_1, \dots, d_h$ in D and therefore it is equivalent with *i*). The second condition is equivalent to say that, for any $d_1, \dots, d_h$ in D ,

$$I(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha)(d_1, \dots, d_h) \\ = \underline{h}(I(\alpha_1)(d_1, \dots, d_h), \dots, I(\alpha_n)(d_1, \dots, d_h)) \rightarrow_i I(\alpha)(d_1, \dots, d_h) \\ \geq p(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha).$$

By the adjunction property, such an inequality is equivalent with *ii*). ■

By Propositions 4.2, 5.2 and Theorem 4.7, we obtain:

THEOREM 5.3. *Let $p : F \rightarrow L$ be a fuzzy program and $s : B_L \rightarrow L$ be a fuzzy set of facts. Then s is a H -model of p if and only if for any fact $\alpha \in \text{Supp}(\text{Gr}(p))$,*

- i) $s(\alpha) \geq \text{Gr}(p)(\alpha)$
and, for any implicative clause $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ in $\text{Supp}(\text{Gr}(p))$,
 ii) $s(\alpha) \geq \text{Gr}(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i \underline{h}(s(\alpha_1), \dots, s(\alpha_n))$.

To calculate the H -models of a fuzzy program, it is very useful the notion of immediate consequence operator (see, also, P. Vojtas [17] and [18]).

DEFINITION 5.4. Let $p : F \rightarrow L$ be a fuzzy program. Then the *immediate consequence* operator associated with p is the operator $T_p : \mathfrak{F}(B_{\mathcal{L}}) \rightarrow \mathfrak{F}(B_{\mathcal{L}})$ defined by setting, for any $s \in \mathfrak{F}(B_{\mathcal{L}})$ and $\alpha \in B_{\mathcal{L}}$,

$$T_1(s)(\alpha) = \text{Gr}(p)(\alpha) \vee s(\alpha), \\ T_2(s)(\alpha) = \text{Sup}\{\text{Gr}(p)(h(\alpha_1, \dots, \alpha_n) \otimes_i \underline{h}(s(\alpha_1), \dots, s(\alpha_n)) \rightarrow_i \alpha) \\ : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(\text{Gr}(p))\}, \\ T_p(s) = T_1(s) \cup T_2(s).$$

The immediate consequence operator enables us to give the following elegant and useful characterization of the H -models of a fuzzy program.

THEOREM 5.5. *Let $p : F \rightarrow L$ be a fuzzy program and $s \in \mathfrak{F}(B_L)$. Then s is an H -model of p if and only if s is a fixed point for T_p .*

PROOF. It is sufficient to apply Theorem 5.3. Indeed, let s be a fixed point for T_p , i.e. assume that $T_1(s) \subseteq s$ and $T_2(s) \subseteq s$. Let α be a fact in $\text{Supp}(Gr(p))$. Then, since $s(\alpha) \geq T_1(s)(\alpha) \geq Gr(p)(\alpha)$, we have that $i)$ is satisfied. Let $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ be a ground implicative clause in $\text{Supp}(Gr(p))$. Then, since $s(\alpha) \geq T_2(s)(\alpha)$, we have that

$$s(\alpha) \geq Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i \underline{h}(s(\alpha_1), \dots, s(\alpha_n))$$

for any $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ in $\text{Supp}(p)$. Hence, $ii)$ is satisfied. This proves that s is a H -model of p . Conversely, assume that s is a H -model of p and therefore that both $i)$ and $ii)$ are satisfied. Then, by $i)$, we have that, for any fact α , $s(\alpha) \geq Gr(p)(\alpha)$ and therefore $s(\alpha) \geq T_1(\alpha)$. Moreover, by $ii)$,

$$s(\alpha) \geq Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i \underline{h}(s(\alpha_1), \dots, s(\alpha_n))$$

and therefore

$$\begin{aligned} s(\alpha) &\geq \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n)) \otimes_i \underline{h}(s(\alpha_1), \dots, s(\alpha_n))) \\ &\quad : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\} \vee s(\alpha) = T_2(s)(\alpha). \end{aligned} \quad \blacksquare$$

6. Calculus of the fuzzy Herbrand models

Any fuzzy program p admits as a trivial H -model the map $s : B_L \rightarrow L$ constantly equal to 1. A more interesting H -model is the *least Herbrand H -model* of p , i.e. a H -model $m_p : B_L \rightarrow L$ of p which is contained in any H -model of p . This since m_p represents, in a sense, the informative content of p . To see how to obtain such a model, we recall some elementary notions in the theory of fixed points in a complete lattice. We say that a family $(s_i)_{i \in I}$ of fuzzy subsets of a set S is *directed* provided that for every s_i and s_j there is s_h such that $s_i \subseteq s_h$ and $s_j \subseteq s_h$. An operator $J : \mathfrak{F}(S) \rightarrow \mathfrak{F}(S)$ is *continuous* provided that, for any directed family $(s_i)_{i \in I}$ of fuzzy subsets of S ,

$$J\left(\bigcup_{i \in I} s_i\right) = \bigcup_{i \in I} J(s_i).$$

A *continuous almost closure operator* is a continuous operator J such that $J(s) \supseteq s$.

PROPOSITION 6.1. Let $J_1 : \mathfrak{F}(S) \rightarrow \mathfrak{F}(S)$ and $J_2 : \mathfrak{F}(S) \rightarrow \mathfrak{F}(S)$ be two continuous operators. Then the union $J = J_1 \cup J_2$ is a continuous operators.

PROOF. Let $(s_j)_{j \in J}$ be a directed family of fuzzy subsets of S and observe that for any $i \in I$ and $j \in I$, there is $h \in I$ such that $s_i \subseteq s_h$ and $s_j \subseteq s_h$, and therefore $J_1(s_i) \subseteq J_1(s_h)$ and $J_2(s_j) \subseteq J_2(s_h)$. Then

$$\begin{aligned}
J\left(\bigcup_{i \in I} s_i\right) &= J_1\left(\bigcup_{i \in I} s_i\right) \cup J_2\left(\bigcup_{i \in I} s_i\right) \\
&= \left(\bigcup_{i \in I} J_1(s_i)\right) \cup \left(\bigcup_{i \in I} J_2(s_i)\right) \\
&= \bigcup \{J_1(s_i) \cup J_2(s_j) : i \in I, j \in I\} \\
&= \bigcup_{h \in I} (J_1(s_h) \cup J_2(s_h)) = \bigcup_{h \in I} J(s_h).
\end{aligned}$$

■

The following theorem summarizes the main properties of the continuous almost closure operators (see, for example, [7]).

THEOREM 6.2. *Let $J : \mathfrak{F}(S) \rightarrow \mathfrak{F}(S)$ be a continuous almost closure operator, then:*

- *the class of fixed points of J is a closure system, i.e. it is closed under arbitrary intersections,*
- *the least fixed point operator $D : \mathfrak{F}(S) \rightarrow \mathfrak{F}(S)$ defined by setting, for any fuzzy subset s ,*

$$D(s) = \bigcap \{s' : J(s') = s', s' \supseteq s\},$$

is a closure operator,

- *for any fuzzy subset s ,*

$$D(s) = \bigcup_{n \in \mathbb{N}} J^n(s). \quad (6.1)$$

We can apply this theorem to the immediate consequence operator.

THEOREM 6.3. *Let p be a positive fuzzy program. Then T_p is a continuous almost closure operator. As a consequence, the class of fuzzy H -models of a positive fuzzy program p is a closure system and the least fuzzy Herbrand model m_p for p is given by:*

$$m_p = \bigcap \{s \in \mathfrak{F}(B_{\Sigma}) : s \text{ is a model of } p\}. \quad (6.2)$$

Moreover,

$$m_p = \bigcup_{n \in \mathbb{N}} T_p^n(\emptyset). \quad (6.3)$$

PROOF. Let T_1 and T_2 be as in Definition 5.4. It is immediate that T_1 satisfies the continuity property. Assume that $(s_j)_{j \in J}$ is a directed family of fuzzy subsets of B_{Σ} . Then

$$\begin{aligned}
T_2(\bigcup_{j \in J} s_j)(\alpha) &= \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i \underline{h}(\text{Sup}_{j \in J} s_j(\alpha_1), \\
&\quad \dots, \text{Sup}_{j \in J} s_j(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\} \\
&= \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i (\text{Sup}_{j(1), \dots, j(n)} h(s_{j(1)}(\alpha_1), \\
&\quad \dots, s_{j(n)}(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\}.
\end{aligned}$$

Now, we observe that, since $(s_j)_{j \in J}$ is directed, given $s_j(1), \dots, s_j(n)$ there is $j \in J$ such that $s_{j(1)} \subseteq s_j, \dots, s_{j(n)} \subseteq s_j$ and therefore

$$\underline{h}(s_{j(1)}(\alpha_1), \dots, s_{j(n)}(\alpha_n)) \leq \underline{h}(s_j(\alpha_1), \dots, s_j(\alpha_n)).$$

Consequently,

$$\begin{aligned}
T_2(\bigcup_{j \in J} s_j)(\alpha) &= \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes (\text{Sup}_{j \in J} \underline{h}(s_j(\alpha_1), \\
&\quad \dots, s_j(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\} \\
&= \text{Sup}_{j \in J} \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes (\underline{h}(s_j(\alpha_1) \\
&\quad \dots, s_j(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\}, \\
&= \text{Sup}_{j \in J} T_2(s_j)(\alpha) = \bigcup_{j \in J} T_2(s_j)(\alpha).
\end{aligned}$$

The remaining part of the theorem is trivial. ■

Equation (6.3) defines an algorithm to calculate, for any fact α , the value $m_p(\alpha)$. Indeed, in accordance with the definition of recursive enumerability for fuzzy sets proposed in [6] and [1], it is easy to show that m_p is a recursively enumerable fuzzy subset of $B_{\mathcal{L}}$. The following theorem shows that the least fuzzy Herbrand model of a positive fuzzy program represents its informative content.

THEOREM 6.4. *Let p be a positive fuzzy program. Then the least fuzzy Herbrand model of p is equal to the fuzzy subset of facts which are logical consequences of p , i.e., for any fact α ,*

$$m_p(\alpha) = Lc(p)(\alpha). \quad (6.4)$$

PROOF. By Theorem 4.6 and (6.2), for any fact α ,

$$Lc(p)(\alpha) = \text{Inf}\{||\alpha||_{\mathcal{J}} : \mathcal{J} \text{ is an } H\text{-model of } p\} = m_p(\alpha). \quad \blacksquare$$

As in the classical case, several difficulties exist for fuzzy programs which are non positive. We confine ourselves only to the very simple case of *hierarchical* fuzzy programs. We call *level mapping* any map $lev : B_{\mathcal{L}} \rightarrow N$ from the Herbrand base to the set of natural numbers. We say that a fuzzy program p is *hierarchical* if a level mapping exists such that for any rule $h(\alpha_1, \dots, \alpha_n) \rightarrow \alpha$ in $\text{Supp}(Gr(p))$, we have that $lev(\alpha) > lev(\alpha_1), \dots, lev(\alpha) > lev(\alpha_n)$. It is no restrictive to assume that the level of the atomic formulas in $\text{Supp}(Gr(p))$ is equal to 1.

PROPOSITION 6.5. *Let p be a hierarchical fuzzy program. Then, for any $\alpha \in B_{\mathcal{L}}$,*

$$\text{lev}(\alpha) \leq n \Rightarrow T_p^{n+1}(\emptyset)(\alpha) = T_p^n(\emptyset)(\alpha). \quad (6.5)$$

Consequently, $\bigcup_{n \in N} T_p^n(\phi)$ is a H -model of p .

PROOF. We will prove (6.5) by induction on n . In the case $n = 1$, we have that α is atomic and that no rule in $\text{Supp}(Gr(p))$ exists whose head is α . Then, $T_2(T_p(\emptyset))(\alpha) = 0$ and $T_2(\emptyset)(\alpha) = 0$. Therefore, since $T_p(\emptyset)(\alpha) = Gr(p)(\alpha)$,

$$\begin{aligned} T_p^2(\emptyset)(\alpha) &= T_p(T_p(\emptyset))(\alpha) = T_1(T_p(\emptyset))(\alpha) \vee T_2(T_p(\emptyset))(\alpha) = T_1(T_p(\emptyset))(\alpha) \\ &= Gr(p)(\alpha) \vee T_p(\emptyset)(\alpha) = T_p(\emptyset)(\alpha). \end{aligned}$$

Assume that $n \neq 1$, that $\text{lev}(\alpha) \leq n$ and that $h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha$ is a rule in $\text{Supp}(Gr(p))$. Then, since $\text{lev}(\alpha_i) < \text{lev}(\alpha) \leq n$, by inductive hypothesis $T_p^n(\emptyset)(\alpha_i) = T_p^{n-1}(\emptyset)(\alpha_i)$. Then,

$$\begin{aligned} T_p^{n+1}(\emptyset)(\alpha) &= Gr(p)(\alpha) \vee T_p^n(\emptyset)(\alpha) \vee \text{Sup}\{Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \\ &\quad \otimes_i \underline{h}(T_p^n(\alpha_1), \dots, T_p^n(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\} \\ &= Gr(p)(\alpha) \vee T_p^n(\emptyset)(\alpha) \vee \text{Sup}\{(Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \\ &\quad \otimes_i \underline{h}(T_p^{n-1}(\alpha_1), \dots, T_p^{n-1}(\alpha_n)) : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\}. \end{aligned}$$

By noticing that $Gr(p)(\alpha) \leq T_p^n(\emptyset)(\alpha)$ and

$$\begin{aligned} &\text{Sup}\{(Gr(p)(h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha) \otimes_i \underline{h}(T_p^{n-1}(\alpha_1), \dots, T_p^{n-1}(\alpha_n)) \\ &\quad : h(\alpha_1, \dots, \alpha_n) \rightarrow_i \alpha \in \text{Supp}(Gr(p))\} \leq T_p^n(\emptyset)(\alpha), \end{aligned}$$

we obtain that $T_p^{n+1}(\emptyset)(\alpha) \leq T_p^n(\emptyset)(\alpha)$ and therefore that

$$T_p^{n+1}(\emptyset)(\alpha) = T_p^n(\emptyset)(\alpha). \quad \blacksquare$$

In the following we say that $\bigcup_{n \in N} T_p^n$ is the canonical H -model of p .

7. Fuzzy control

Fuzzy control is perhaps the main success of fuzzy set theory. Recall that the aim of classical control theory is to individuate a function $\underline{f} : X \rightarrow Y$ such that $\underline{f}(x)$ gives the correct control action given the input x . To do this, the starting point is a general theory about the phenomenon under consideration. From this theory we obtain some differential equations and then $\underline{f}$ is obtained as a solution of these equations. The paradigm used in fuzzy control theory, as devised by Zadeh in [19], [20] and by Mamdani in [12], is totally different. Indeed, in fuzzy control one tries to obtain $\underline{f}$ from the verbal information given by an expert on the control under consideration (imagine the expert as a cleaver 'old hand' with no theoretical knowledge or

mathematical competence). The expert expresses its experience by a system of IF-THEN rules like

$$\begin{aligned}
 &\text{If } x \text{ is Little, then I suggest to put } y \text{ Slow,} \\
 &\text{If } x \text{ is Small, then I suggest to put } y \text{ Fast,} \\
 &\text{If } x \text{ is Medium, then I suggest to put } y \text{ Moderate,} \\
 &\text{If } x \text{ is Big, then I suggest to put } y \text{ Veryfast,} \\
 &\text{If } x \text{ is Verybig, then I suggest to put } y \text{ Moderate.}
 \end{aligned} \tag{7.1}$$

Successively, fuzzy control theory interprets such an information, qualitative in nature, by translating it into a function $f' : X \rightarrow Y$ in such a way that f' can be considered an adequate control function. This is done by suitable procedures based on the interpretation the words *Little*, *Slow*, ... by suitable fuzzy quantities, or fuzzy granules, $little : X \rightarrow [0, 1]$, $slow : Y \rightarrow [0, 1]$, By proceeding in a general way, consider the **IF-THEN** system:

$$\begin{aligned}
 &\text{IF } x \text{ is } A_1 \text{ THEN } y \text{ is } B_1, \\
 &\dots \\
 &\text{IF } x \text{ is } A_n \text{ THEN } y \text{ is } B_n.
 \end{aligned} \tag{7.2}$$

Then, in fuzzy control such a system can be interpreted in two different ways. In the *conjunction-based* fuzzy control we consider the following procedure where $\otimes$ is a binary operation able to interpret the conjunction.

- **Step 1.** A tentative interpretation of the constants $A_1, \dots, A_n, B_1, \dots, B_n$ by suitable "fuzzy quantities" $a_1 : X \rightarrow [0, 1], \dots, a_n : X \rightarrow [0, 1], b_1 : Y \rightarrow [0, 1], b_n : Y \rightarrow [0, 1]$ is proposed.
- **Step 2.** Each rule "IF x is A_i THEN y is B_i " is associated with the fuzzy point $a_i \times b_i$ obtained as the $\otimes$ -Cartesian product of the fuzzy quantities a_i and b_i .
- **Step 3.** The whole system of rules is associated with the fuzzy function $f : X \times Y \rightarrow [0, 1]$ defined as an union of these fuzzy points, i.e.

$$f = \bigcup_{i=1, \dots, n} (a_i \times b_i). \tag{7.3}$$

- **Step 4.** A suitable *defuzzification process* associates the fuzzy function f with a crisp function f' . Usually, this is obtained by the centroid method, where we set, for every $r \in X$,

$$f'(r) = \frac{\int_Y y \cdot f(r, y) dy}{\int_Y f(r, y) dy}$$

- **Step 5.** If f' is satisfactory (it is sufficiently close to f), then the procedure terminates, otherwise the interpretation of the expert's words is chanced in accordance with a suitable strategy. After this, one goes back to Step 2.

Notice that the basic feature of such a procedure is the tuning process, where the interpretation of the constants $A_1, \dots, A_n, B_1, \dots, B_n$ is changed until we obtain a satisfactory interpretation. By referring to the system (7.1), set $X = [0, 10], Y = [0, 5]$ and interpret the constants *Little, Small, Medium, Big, Verybig, Slow, Fast, Moderate, Veryfast* by suitable triangle functions. Also, consider the triangular norm of the minimum. Then, Figure 1 represents both the corresponding fuzzy function f and the crisp function f' .

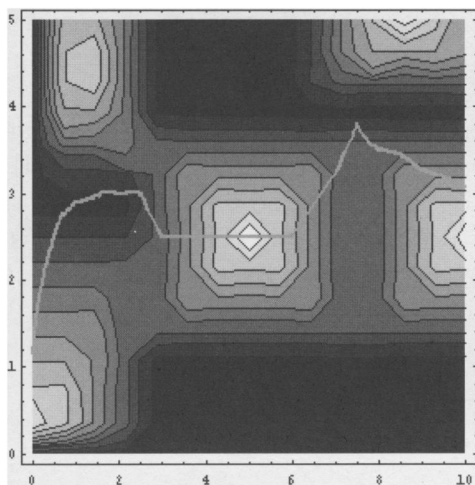


Figure 1.

In the *implication – based* procedure (see, for example, [11]), instead of steps 2 and 3 we have:

- **Step 2’.** Each rule “IF x is A_i THEN y is B_i ” is associated with the fuzzy relation $a_i \rightarrow b_i$ defined by setting, $(a_i \rightarrow b_i)(x, y) = a_i(x) \rightarrow b_i(y)$ for any $x \in X$ and $y \in Y$, and where $\rightarrow$ is a binary operator in $[0, 1]$ able to interpret the implication.
- **Step 3’.** The whole system of rules is associated with the fuzzy function:

$$g = \bigcap_{i=1, \dots, n} (a_i \rightarrow b_i). \quad (7.4)$$

8. Fuzzy control as a chapter of fuzzy logic programming

In spite of the great success of fuzzy control, there is no convincing justification of the applied techniques. Obviously, the first temptation is to interpret the IF-THEN structure of the rules as a logical implication. As an example, we could interpret a rule as “If x is *Little* then y is *Slow*” by the first order formula $Little(x) \rightarrow Slow(y)$. Unfortunately, this cannot be done in a so

direct way. Indeed assume that r is an input such that $Little(r)$ is false. Then the formula $Little(r) \rightarrow Slow(t)$ is true for any t in Y and therefore any $t \in Y$ gives a correct control. As a consequence of these considerations, in [8] we sketched a logical approach to fuzzy control based on the idea that, given the input r , and a possible output t , the number $f(r, t)$ is the truth degree of the claim " t is a good answer given r ". Namely, we show that f is the interpretation of a vague predicate *Good* in the least fuzzy Herbrand model of a suitable fuzzy program. To illustrate this idea, we consider a multivalued logic whose language $\mathcal{L}$ contains a binary relation symbol *Good* and unary predicate symbols $A_1, \dots, A_n, B_1, \dots, B_n$ corresponding to the fuzzy quantities in (7.2). Also, $\mathcal{L}$ contains each element r in X and t in Y as a constant. The predicates are interpreted in the complete lattice $[0, 1]$ equipped with a triangular norm $\otimes$ and the map $\smile: [0, 1] \rightarrow [0, 1]$ defined by setting $\smile(x) = 1 - x$. These operations are devoted to interpret the conjunction and the negation connectives, respectively. The disjunction is interpreted by the co-norm $\oplus$ defined by setting $x \oplus y = \smile(\smile x \otimes \smile y)$ and the implication by the operation $\rightarrow$ defined by setting $x \rightarrow y = (\smile x) \oplus y$. Then, we translate the system (7.2) into a fuzzy program $p: F \rightarrow [0, 1]$ defined by setting:

$$\left\{ \begin{array}{ll} A_1(x) \wedge B_1(y) \rightarrow Good(x, y) & [1] \\ \dots & \\ A_n(x) \wedge B_n(y) \rightarrow Good(x, y) & [1] \\ A_1(r) & [a_1(r)] \\ \dots & \dots \\ A_n(r) & [a_n(r)] \\ B_1(t) & [b_1(t)] \\ \dots & \dots \\ B_n(t) & [b_n(t)], \end{array} \right. \quad (8.1)$$

where r varies in X and t in Y . More precisely, p is defined by setting

$$p(\alpha) = \left\{ \begin{array}{ll} 1 & \text{if } \alpha \text{ is the clause } A_i(x) \wedge B_i(y) \rightarrow Good(x, y), \\ a_i(r) & \text{if } \alpha \text{ is the fact } A_i(r), r \in X \\ b_i(t) & \text{if } \alpha \text{ is the fact } B_i(t), t \in Y \\ 0 & \text{otherwise.} \end{array} \right.$$

Observe that $A_1, \dots, A_n, B_1, \dots, B_n$ are considered as unary predicate symbols and not constants for fuzzy quantities as in the system (7.1). Observe also

that, while $p(A_i(r)) = a_i(r)$ for any $r \in X$, we have set $p(A_i(t)) = 0$ for any $t \in Y$, and that while $p(B_i(t)) = b_i(t)$ for any $t \in Y$, we have set $p(A_i(r)) = 0$ for any $r \in X$ (it is not restrictive to assume that $X \cup Y = \emptyset$). So, the Herbrand universe is the disjoint union of two sets. A better alternative should be to extend the results of this paper into a two sorted logic, one type for the elements in X and one type for the elements in Y .

The following theorem shows that the calculus of the fuzzy function f obtained by the conjunction-procedure is equivalent to the calculus of the least fuzzy Herbrand model of such a fuzzy program.

THEOREM 8.1. *Consider the fuzzy control system (7.2), let f be the associated fuzzy function by the conjunction-based procedure and let p be the fuzzy program given by (8.1). Then, for any $r \in X$ and $t \in Y$,*

$$f(r, t) = m_p(\text{Good}(r, t)),$$

i.e., f is the extension of the predicate Good in the least fuzzy Herbrand model of p .

PROOF. We observe only that, due to the one-level structure of the program p , $m_p = T_p^2(\emptyset)$. ■

Likewise, in order to give a logical interpretation of implication-based fuzzy control, we consider a new binary predicate symbol Bad and the following hierarchical fuzzy program p :

$$\left\{ \begin{array}{ll} A_1(x) \wedge \neg B_1(y) \rightarrow \text{Bad}(x, y) & [1] \\ \dots & \\ A_n(x) \wedge \neg B_n(y) \rightarrow \text{Bad}(x, y) & [1] \\ A_1(r) & [a_1(r)] \\ \dots & \dots \\ A_n(r) & [a_n(r)] \\ B_1(t) & [b_1(t)] \\ \dots & \dots \\ B_n(t) & [b_n(t)], \end{array} \right. \quad (8.2)$$

where $r \in X$ and $t \in Y$. The following theorem shows that the calculus of the fuzzy function f obtained by the implication-procedure is equivalent to the calculus of the least fuzzy Herbrand model of such a fuzzy program.

THEOREM 8.2. *Consider the fuzzy control system (7.2), let g be the fuzzy function obtained by the implication-based procedure and let p be the fuzzy program defined by (8.2). Then, for any $r \in X$ and $t \in Y$,*

$$g(r, t) = \smile m_p(\text{Bad}(r, t)),$$

i.e., g is the extension of the predicate $\neg \text{Bad}$ in the canonical Herbrand model of p .

PROOF. It is immediate that

$$m_p(\text{Bad}(r, t)) = \text{Max}_{i=1, \dots, n} \{a_i(r) \otimes \smile b_i(t)\}$$

and therefore that,

$$\begin{aligned} \smile m_p(\text{Bad}(r, t)) &= \text{Min}_{i=1, \dots, n} (\smile a_i(r) \oplus b_i(t)) \\ &= \bigcap_{i=1, \dots, n} (a_i \rightarrow b_i)(r, t) = g(r, t). \end{aligned}$$

■

Theorems 8.1 and 8.2 enable us to emphasize the different meaning of the two procedures (see, also, [3]). Indeed, while conjunction-based procedure is useful to give positive information (given r , the fuzzy subset $\text{good}(r, y)$ of outputs that we consider good), the implication-based procedure is useful to manage negative information (given r , the fuzzy subset $\text{bad}(r, y)$ of outputs that we suggest to avoid). This gives the possibility of considering both the approaches at the same time. In fact we can consider a hierarchical fuzzy program containing both rules for the predicate *Good* and rules for the predicate *Bad*. As an example, we can consider the fuzzy program obtained by adding at the fuzzy program (8.1), the rule

$$(\neg \text{Big}(x)) \wedge (\text{Fast}(y) \vee \text{Verifast}(y)) \rightarrow \text{Bad}(x, y) \quad [1]$$

able to define *Bad* and the rule

$$\text{Good}(x, y) \wedge \neg \text{Bad}(x, y) \rightarrow \text{Optimum}(x, y) \quad [1]$$

to compose the two different kinds of information. In such a case the fuzzy function we have to consider is the extension of the predicate *Optimum* and not to the extension of the predicate *Good* in the successive defuzzification process, obviously. In Figure 2 we represent both such a fuzzy function and the corresponding crisp function f' obtained by the defuzzification process.

Notice that such an approach to fuzzy control is related with the approach sketched by P. Hájek in [10]. In fact, Hájek defined the predicate *Mamd*(x, y) by the axiom

$$\text{Mamd}(x, y) \leftrightarrow (A_1(x) \wedge B_1(y)) \vee \dots \vee (A_1(x) \wedge B_1(y)).$$

It is immediate that in any H -model in which such an axiom is satisfied, the extension of the predicate *Mamd* is the fuzzy relation defined in fuzzy control. Then, while we associate any IF-THEN system with a fuzzy program p , Hájek refers to the completion of p , in a sense.

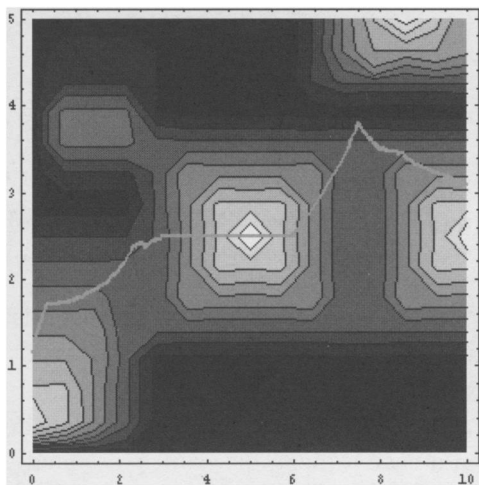


Figure 2.

9. Possible future researches on fuzzy control

As a matter of fact the proposed immersion of fuzzy control into fuzzy logic programming uses only a very small piece of this theory. Then, we can try to extend fuzzy control taking in account the whole potentialities of fuzzy logic programming. For example, we can indicate the following research directions.

9.1. Learning the rules. In defining the fuzzy programs (8.1), we can assign to a rule $A_i(x) \wedge B_i(y) \rightarrow Good(x, y)$ a weight λ_i different from 1 as a measure of the corresponding "degree of confidence" of the expert in such a rule. Then, the interpretation f of $Good$ in the resulting canonical Herbrand model m_p is obtained by the expression

$$f = \bigcup_{i=1, \dots, n} (\lambda_i \otimes (a_i \times b_i)).$$

(9.1)

This means that, in the learning process, we can modify both the interpretations of the vague predicates and the parameters $\lambda_1, \dots, \lambda_n$. Obviously, it is also possible that in such a process a parameter λ_i becomes equal to zero and therefore that the i -rule is deleted.

9.2. Further connectives and linguistic modifiers. We can consider a fuzzy logic with several kind of implication and further logical connectives in $[0,1]$. For instance, we can consider linguistic modifiers as "*Clearly*" and "*Vaguely*" interpreted by the functions $cl : [0, 1] \rightarrow [0, 1]$ and $vag : [0, 1] \rightarrow$

$[0, 1]$ defined by setting $cl(x) = x^2$ and $vag(x) = x^{0.5}$, respectively. Then we can consider a fuzzy control able to manage rules as

$$\begin{aligned} Vaguely(Little(x)) \wedge Fast(y) &\rightarrow Good(x, y), \\ Little(x) \wedge Clearly(Slow(y)) &\rightarrow Good(x, y) \end{aligned}$$

where $Vaguely(Little(x))$ is interpreted by the fuzzy subset $little(x)^{0.5}$ and $Clearly(Slow(y))$ is interpreted by the fuzzy set $slow(y)^2$.

9.3. Boolean algebras, bilattices and further structure for fuzzy control. We can imagine a fuzzy control based on a logic programming with a valuation structure L different from the interval $[0, 1]$. For example, we can assume that S is a set of experts and that L is the power set Boolean algebra $\wp(S)$. In such a case a fuzzy program p associates any clause α with the set $p(\alpha)$ of experts whose opinion is that α is valid. The interpretation of a predicate symbol A_i is a map $s : X \rightarrow \wp(S)$ associating any element $x \in X$ with the set $s(x)$ of experts whose opinion is that A_i is satisfied by x . The same holds for the predicates B_i . In such a case if $f : X \times Y \rightarrow \wp(S)$ is the resulting fuzzy function, then $f(x, y)$ is the set of experts whose opinion is that y is the correct control answer given x . A similar interpretation is obtained by assuming that S is the set of past cases in a statistical data-base. Also, in accordance with [4], should be very interesting to assume that L is a bilattice based on $[0, 1] \times [0, 1]$. The resulting fuzzy control should be able to manage both positive and negative information.

9.4 Rule chaining. The power of logic programming is mainly based on recursion and on many-steps deductions. Instead fuzzy control usually considers one-step deductions. Then, our translation suggests a fuzzy control able to define predicates which, in turn, are used to define new predicates and so on. A simple example is furnished in the previous section in defining *Optimum* from *Good* and *Bad*. To give a further example, assume that an element r in X exists such that $a_1(r) = \dots = a_n(r) = 0$. Then $f(r, t) = 0$ for any $t \in Y$. This means that the proposed rules are not able to suggest a correct control given r . In such a case we can add to the rules used to define *Good* the default rule

$$Undefined(x) \wedge Bydefault(y) \rightarrow Good(x, y),$$

where, in turn, the predicate *Undefined* is defined by the rule:

$$\neg A_1(x) \wedge \dots \wedge \neg A_n(x) \rightarrow Undefined(x)$$

and the meaning of *Bydefault(y)* is defined by further suitable rules.

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